

DAPPC - LAB 2

SOM Results: Hits Maps, Neuron Clusters & Dendrograms

4 datasets × 3 SOM sizes = 12 SOMs

Datasets: KNN Global · KNN By-Class · Clustering Global · Clustering By-Class

SOM sizes: 10×10 · 12×12 · 14×14

Clustering: complete linkage, cityblock distance | Best k: min max intra-cluster variability

★ marks the selected SOM for each dataset

KNN Global Imputation

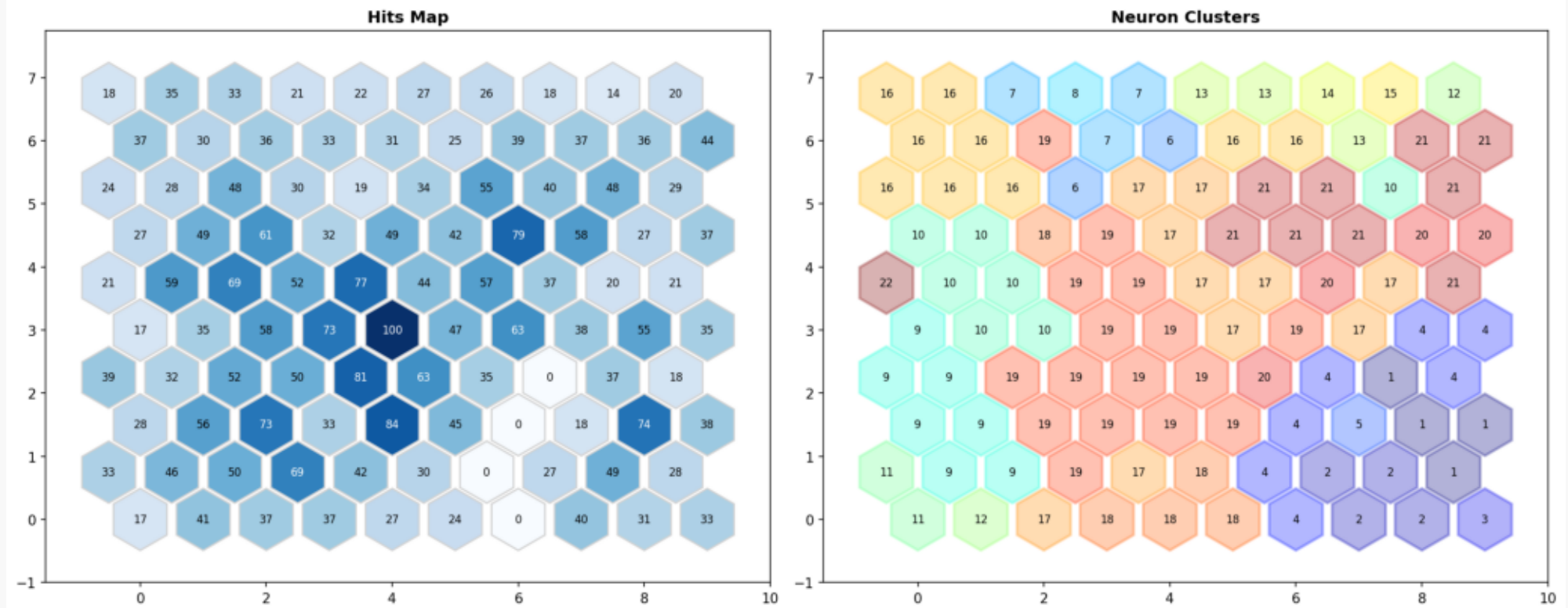
Dataset key: knn_global

★ Selected SOM: 12x12 | best k = 22 | max intra-var = 0.7566

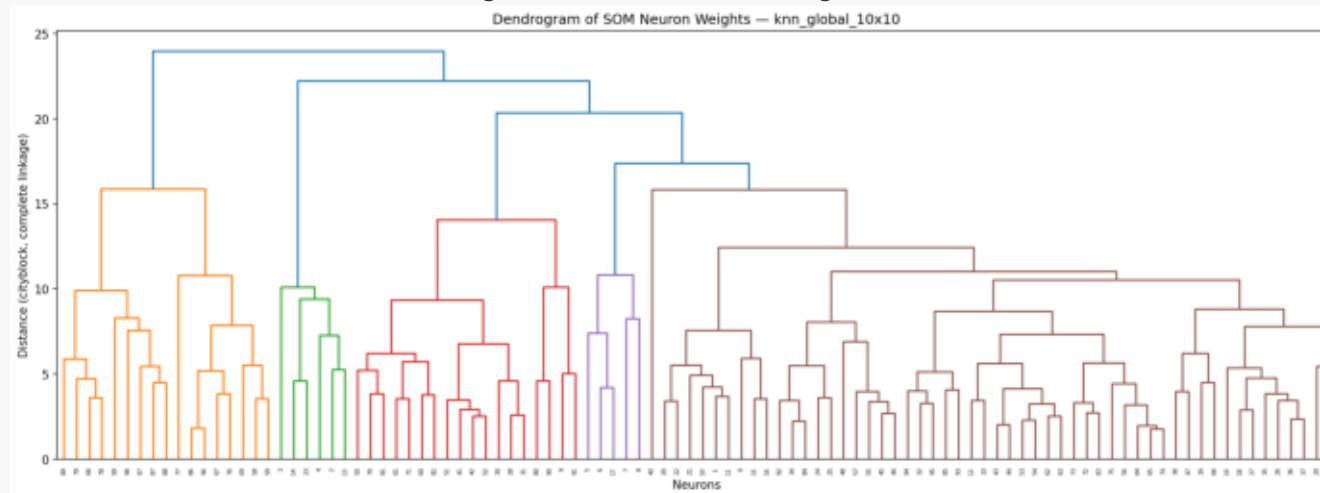
KNN Global Imputation — SOM 10x10 — best k = 22 — max intra-var = 0.8272

Hits Map (left) & Neuron Clusters (right)

SOM_knn_global_10x10_k22



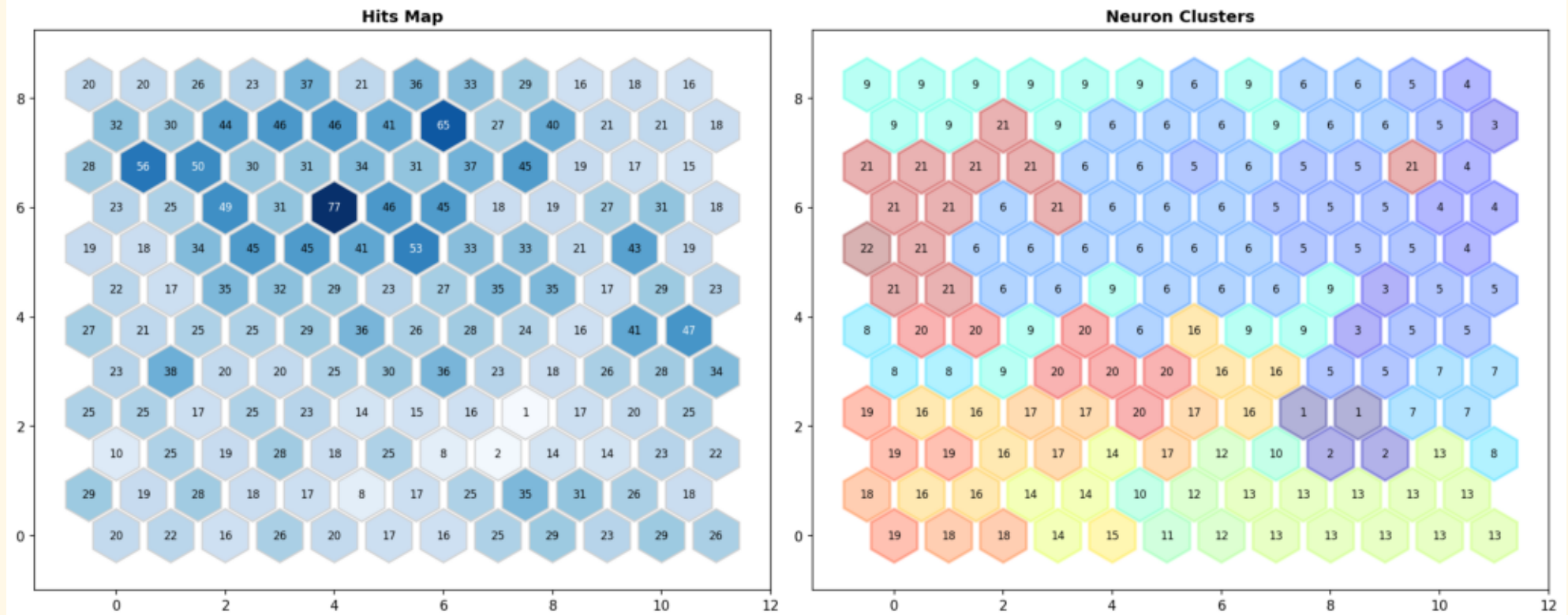
Dendrogram of SOM Neuron Weight Vectors



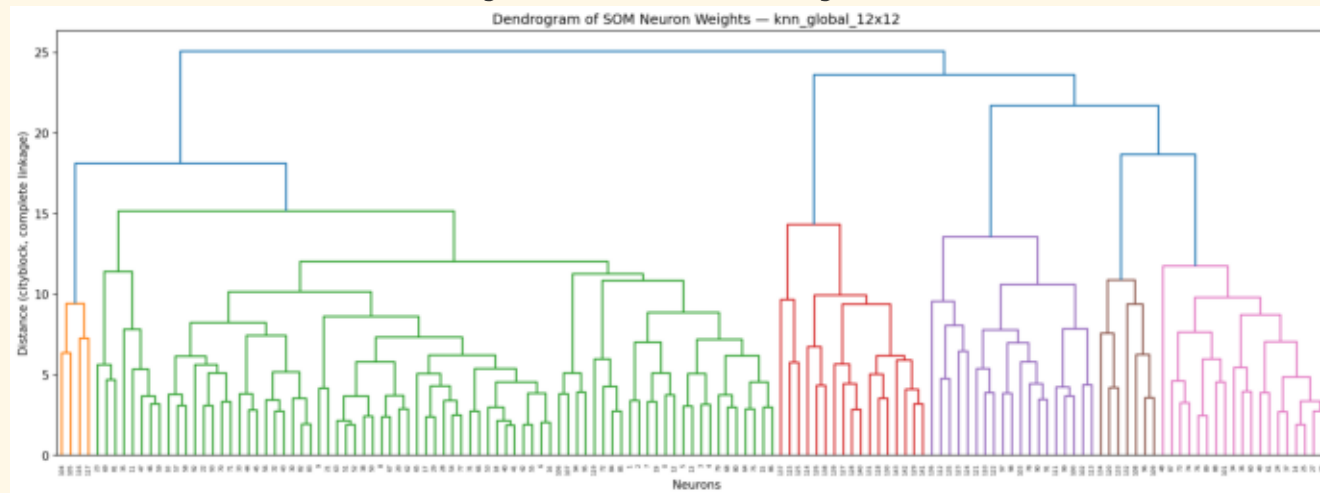
Hits Map (left) & Neuron Clusters (right)

★ SELECTED ★

SOM_knn_global_12x12_k22



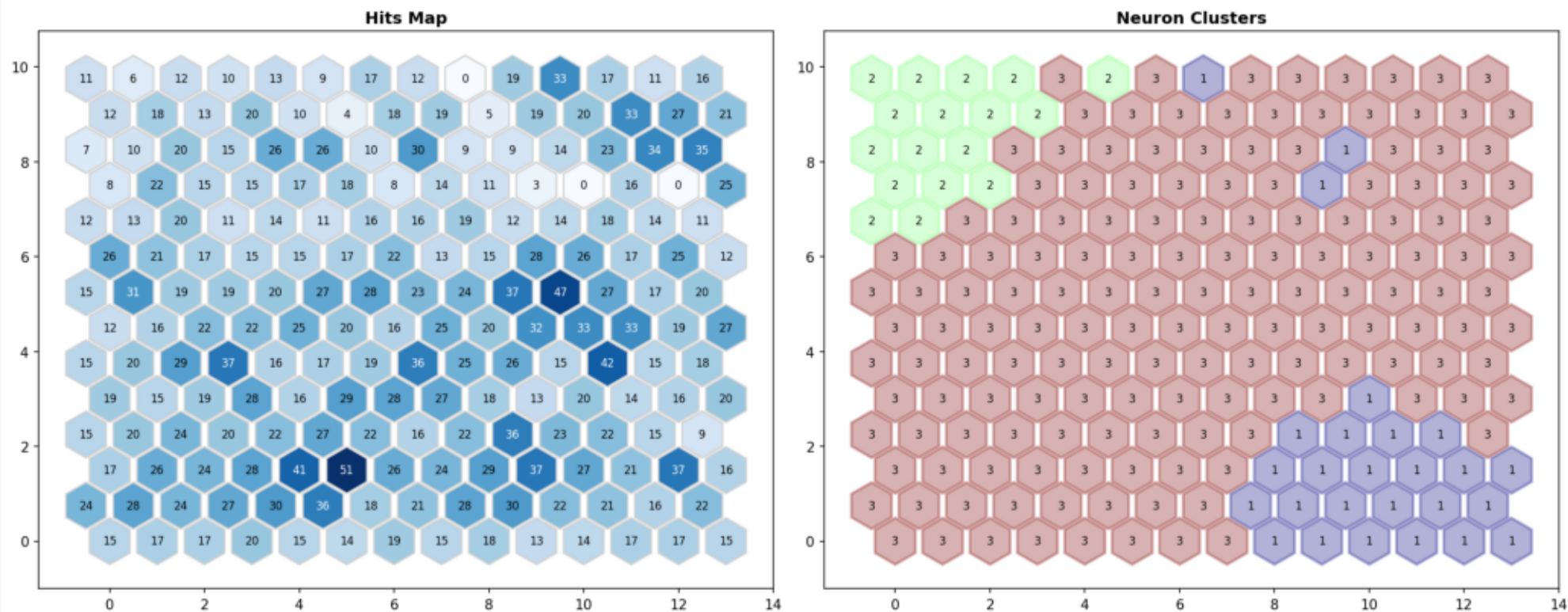
Dendrogram of SOM Neuron Weight Vectors



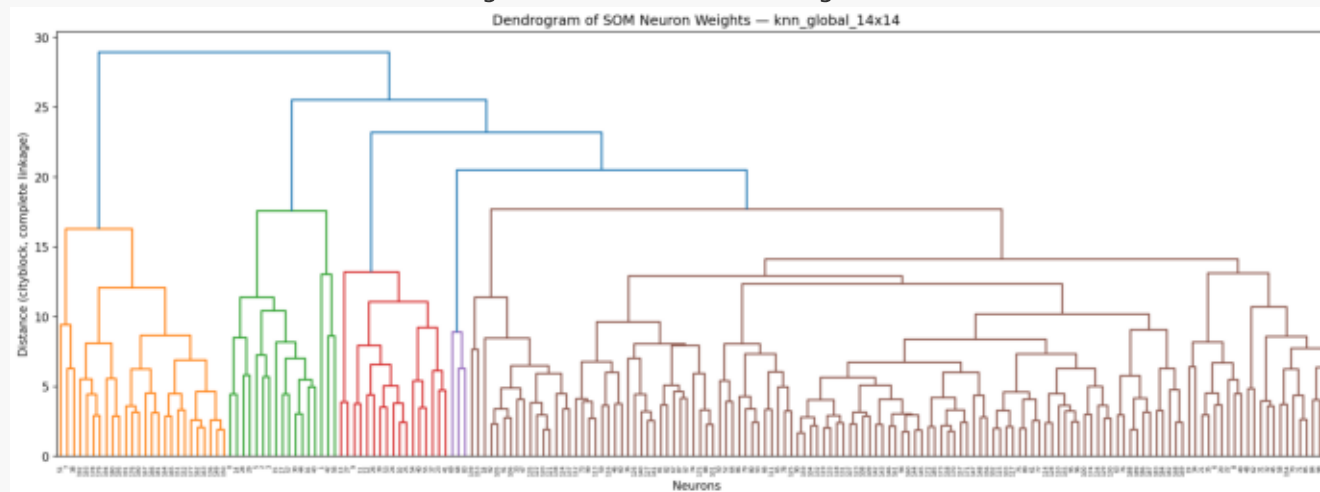
KNN Global Imputation — SOM 14x14 — best k = 3 — max intra-var = 0.9353

Hits Map (left) & Neuron Clusters (right)

SOM_knn_global_14x14_k3



Dendrogram of SOM Neuron Weight Vectors



KNN By-Class Imputation

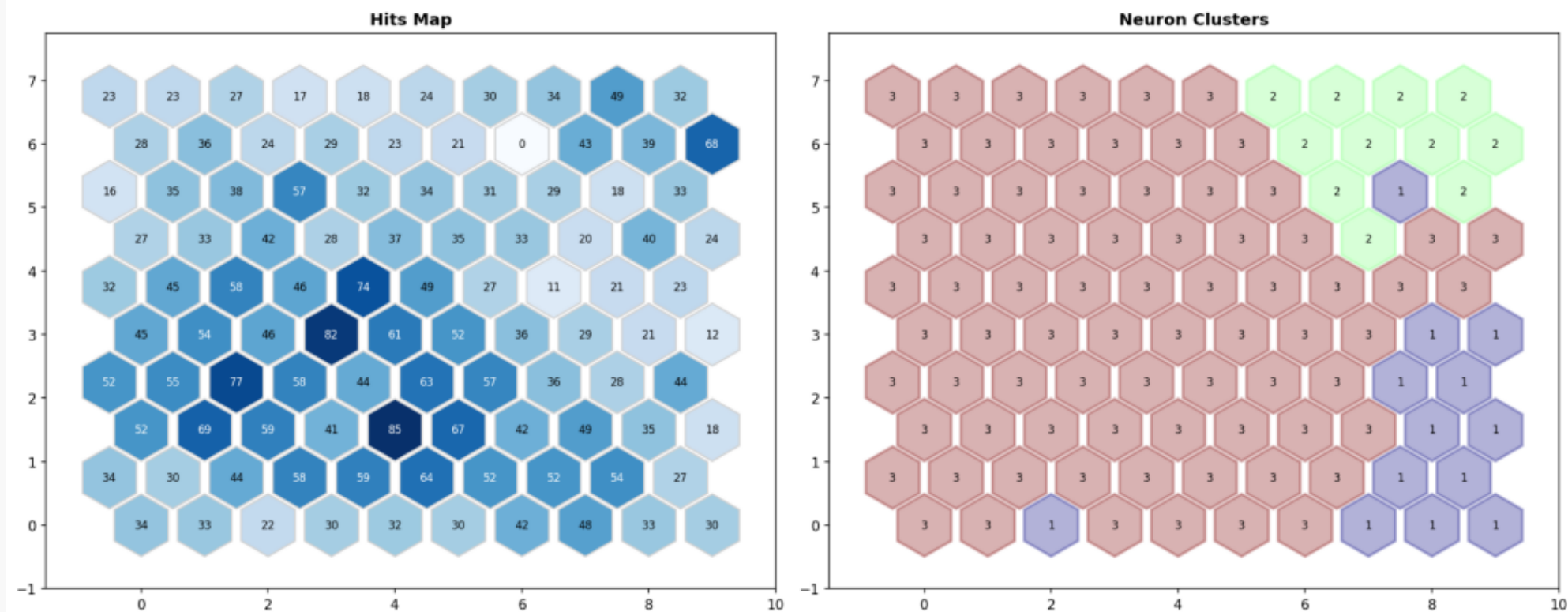
Dataset key: knn_by_class

★ Selected SOM: 14x14 | best k = 5 | max intra-var = 0.8184

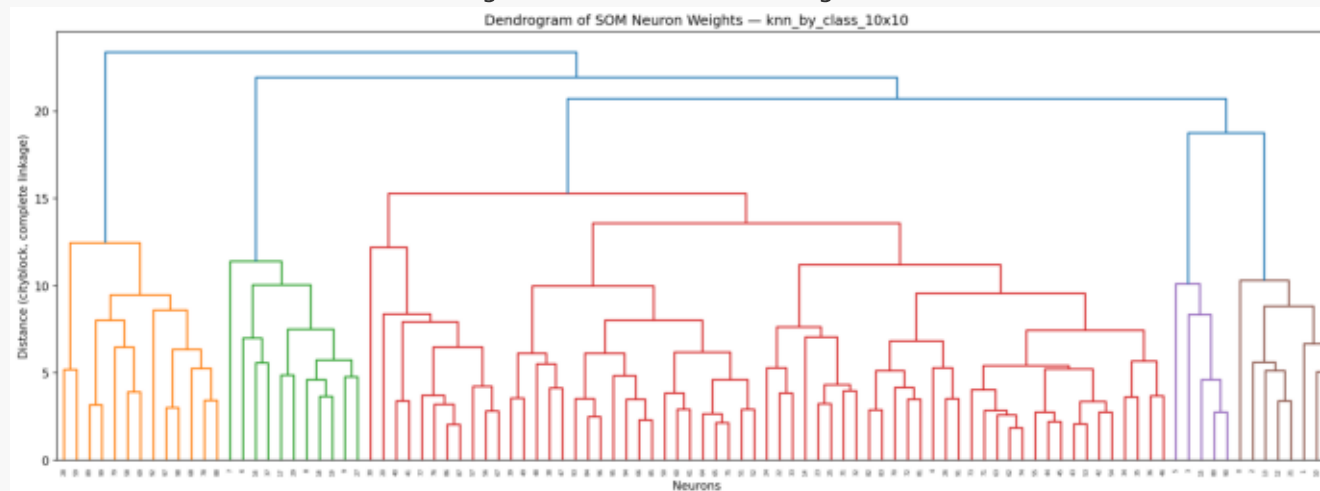
KNN By-Class Imputation — SOM 10x10 — best k = 3 — max intra-var = 0.8265

Hits Map (left) & Neuron Clusters (right)

SOM_knn_by_class_10x10_k3



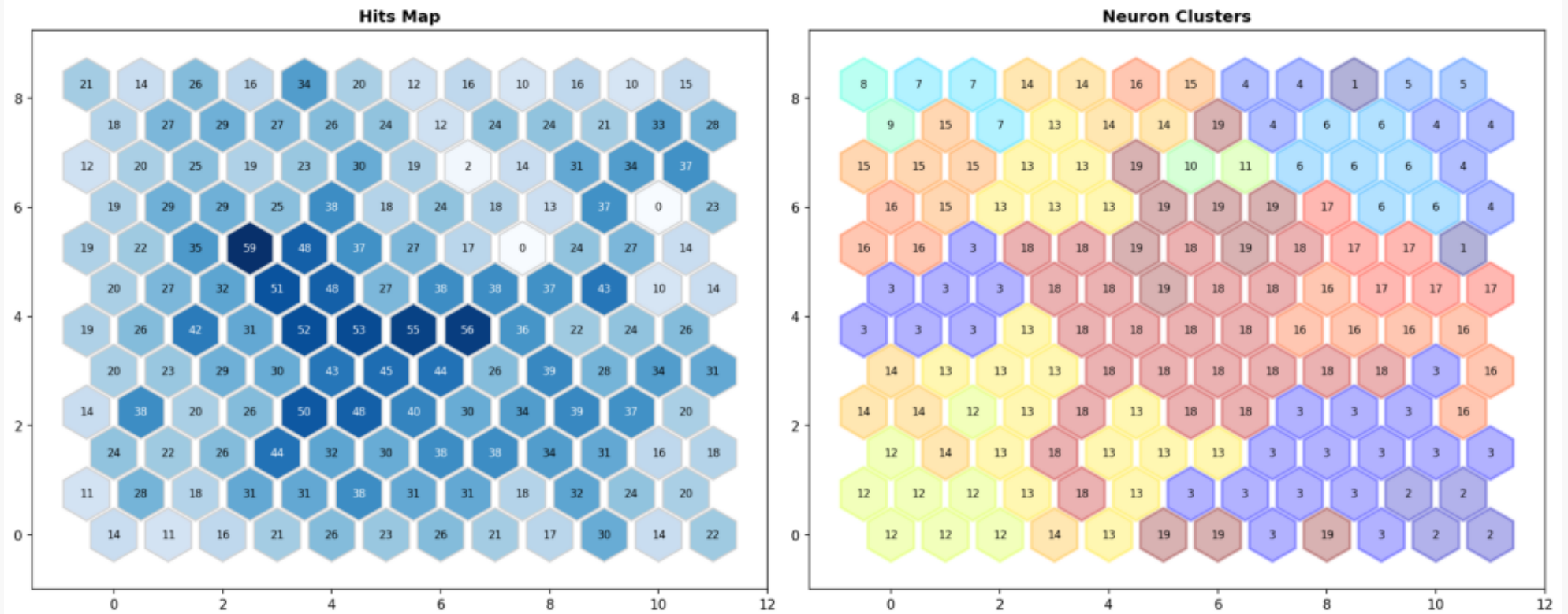
Dendrogram of SOM Neuron Weight Vectors



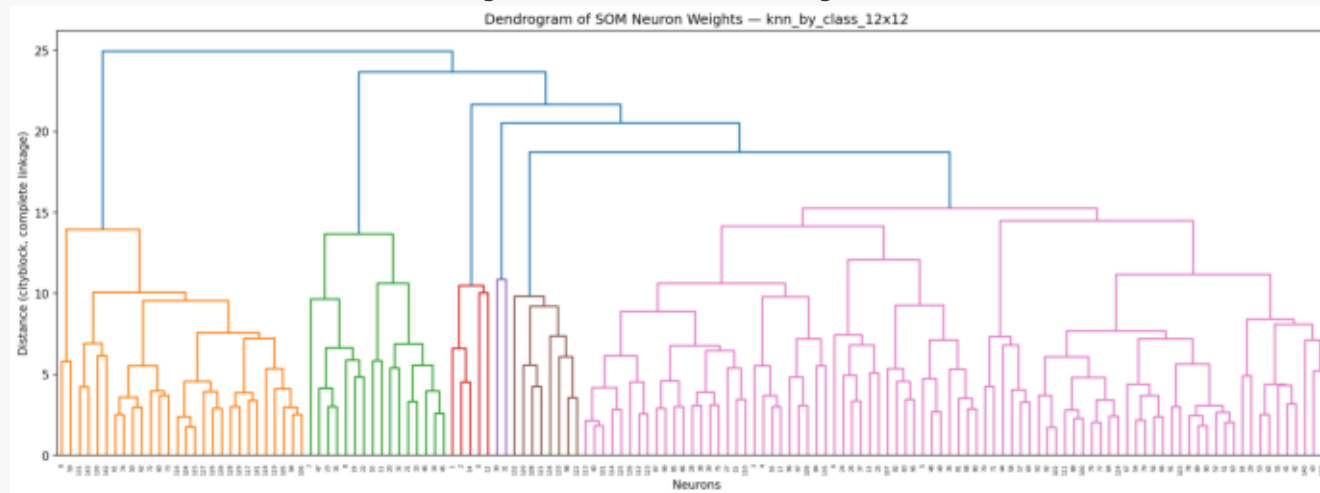
KNN By-Class Imputation — SOM 12x12 — best k = 19 — max intra-var = 0.8272

Hits Map (left) & Neuron Clusters (right)

SOM_knn_by_class_12x12_k19



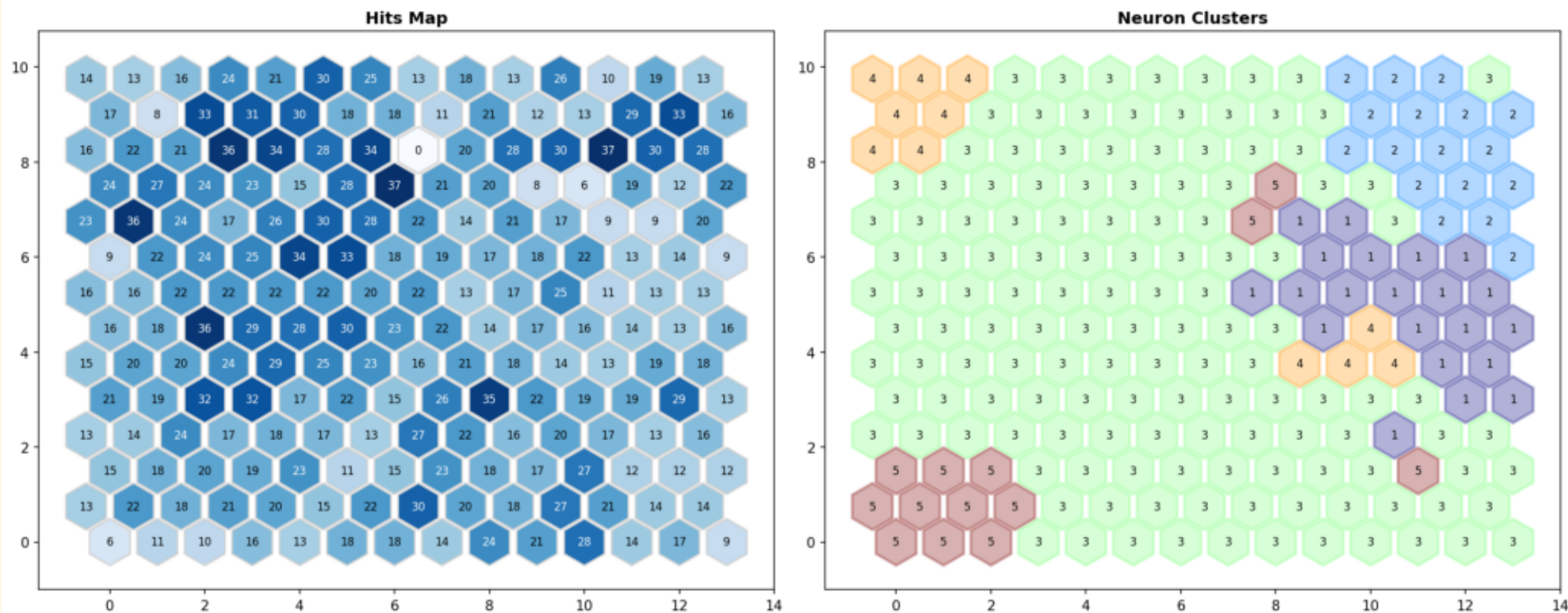
Dendrogram of SOM Neuron Weight Vectors



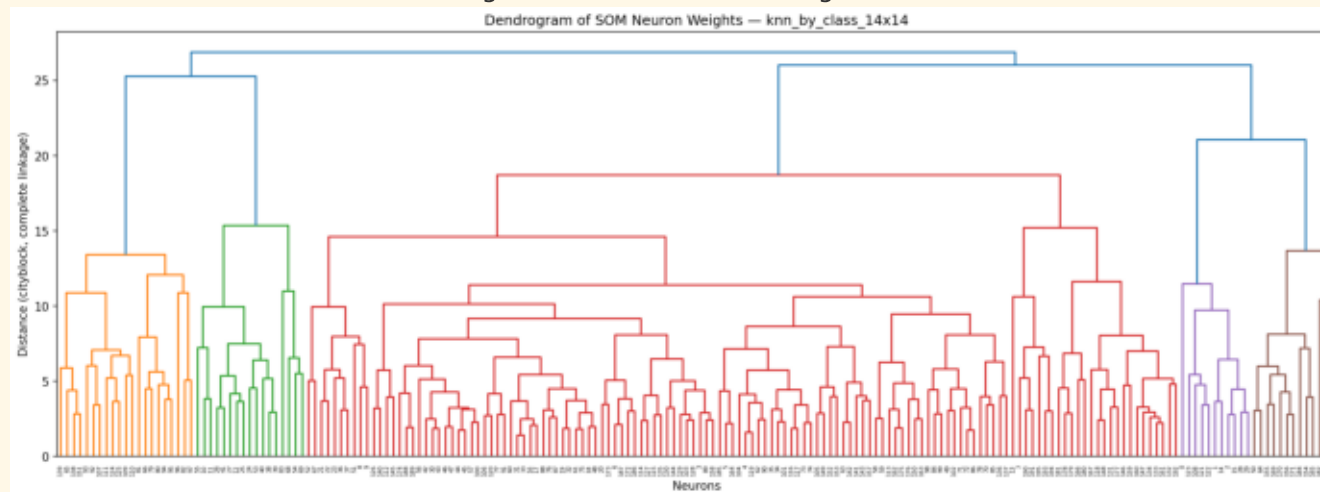
Hits Map (left) & Neuron Clusters (right)

★ **SELECTED** ★

SOM_knn_by_class_14x14_k5



Dendrogram of SOM Neuron Weight Vectors



Clustering Global Imputation

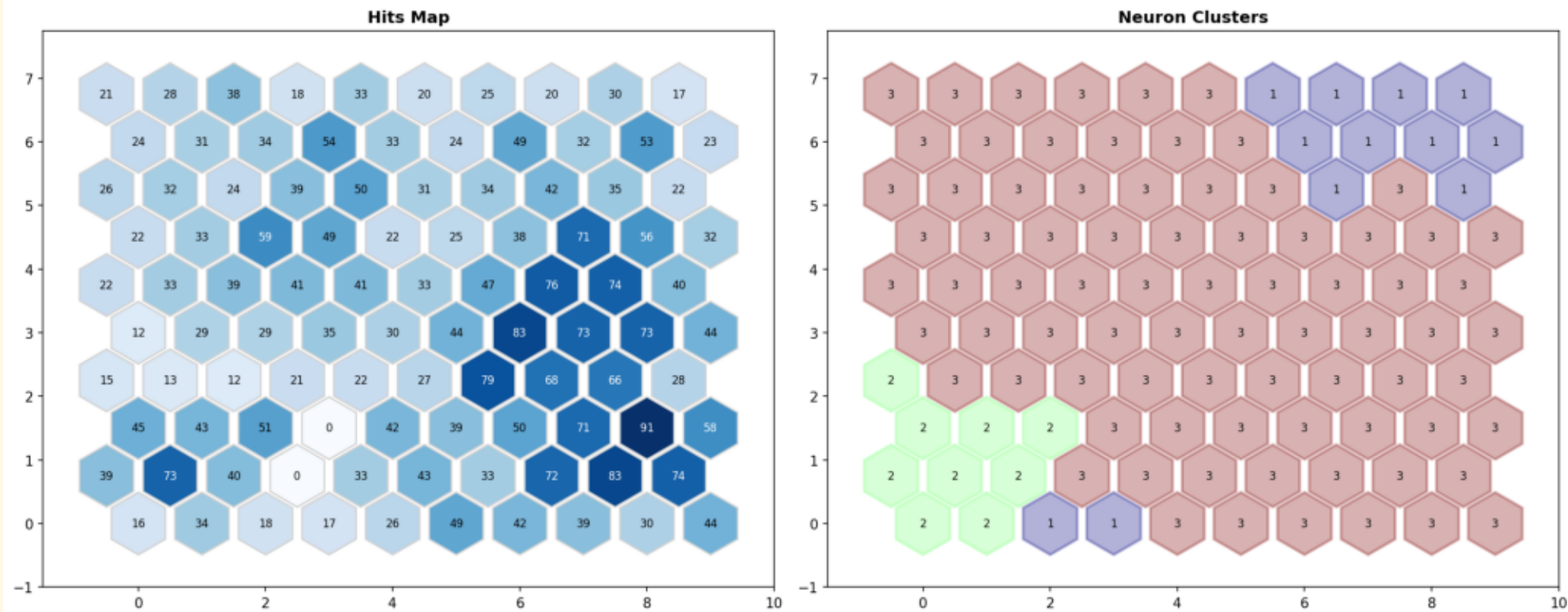
Dataset key: clust_global

★ Selected SOM: 10x10 | best k = 3 | max intra-var = 0.8369

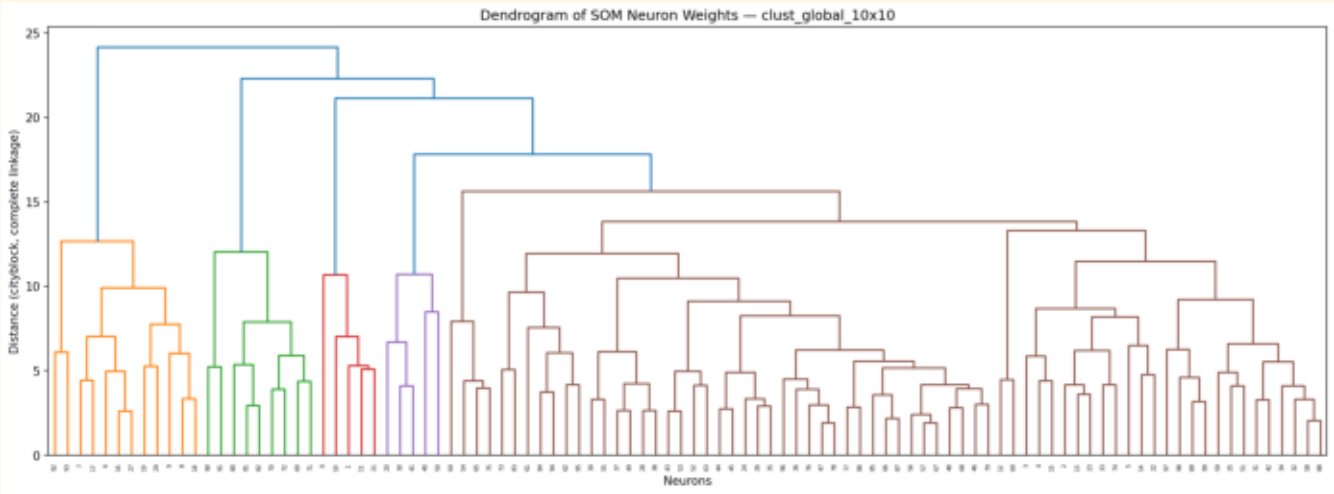
Hits Map (left) & Neuron Clusters (right)

★ SELECTED ★

SOM_clust_global_10x10_k3



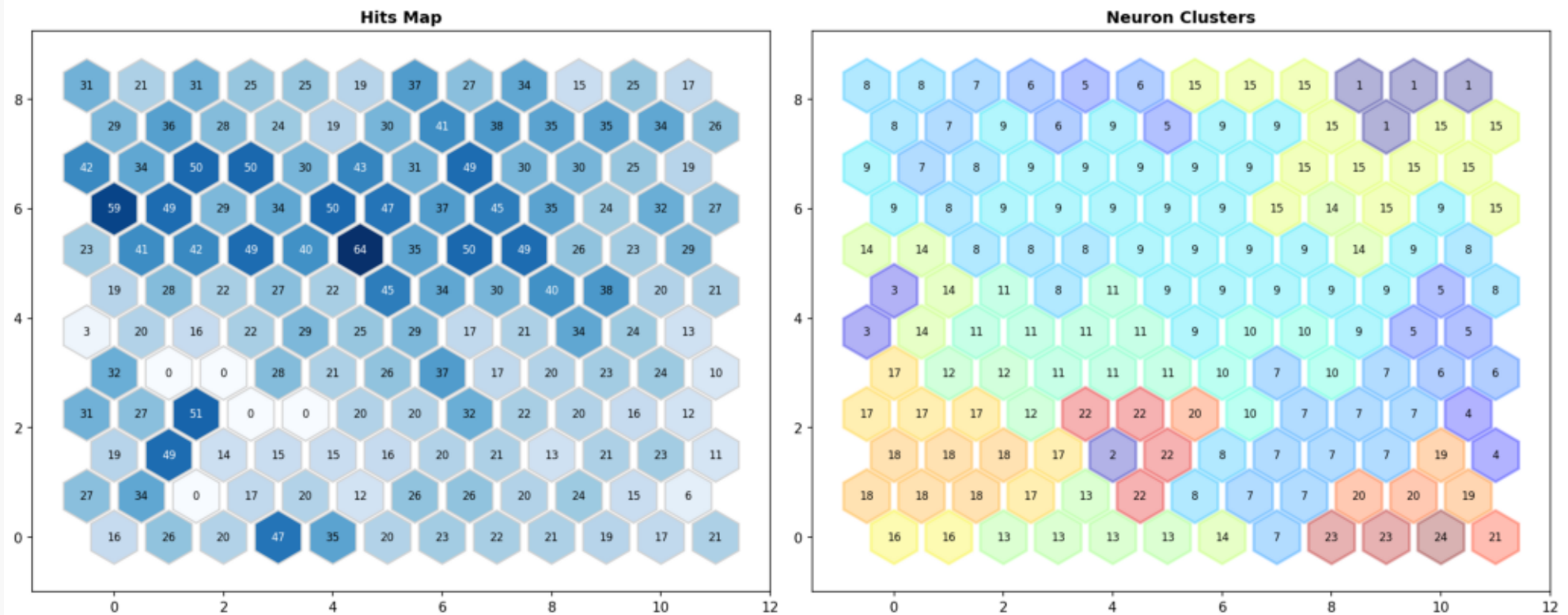
Dendrogram of SOM Neuron Weight Vectors



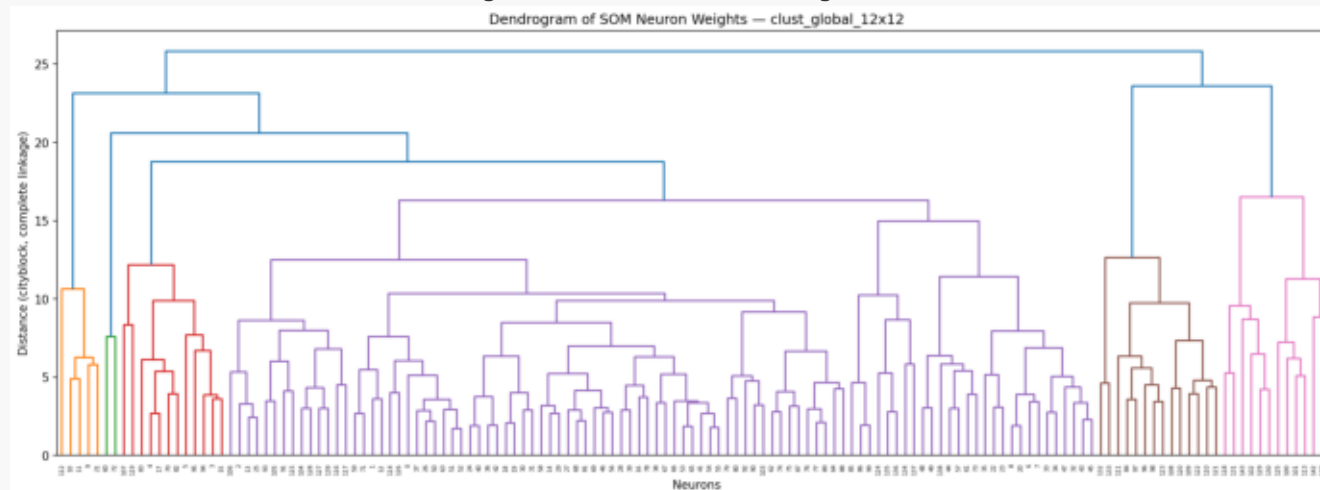
Clustering Global Imputation — SOM 12x12 — best k = 24 — max intra-var = 0.8557

Hits Map (left) & Neuron Clusters (right)

SOM_clust_global_12x12_k24



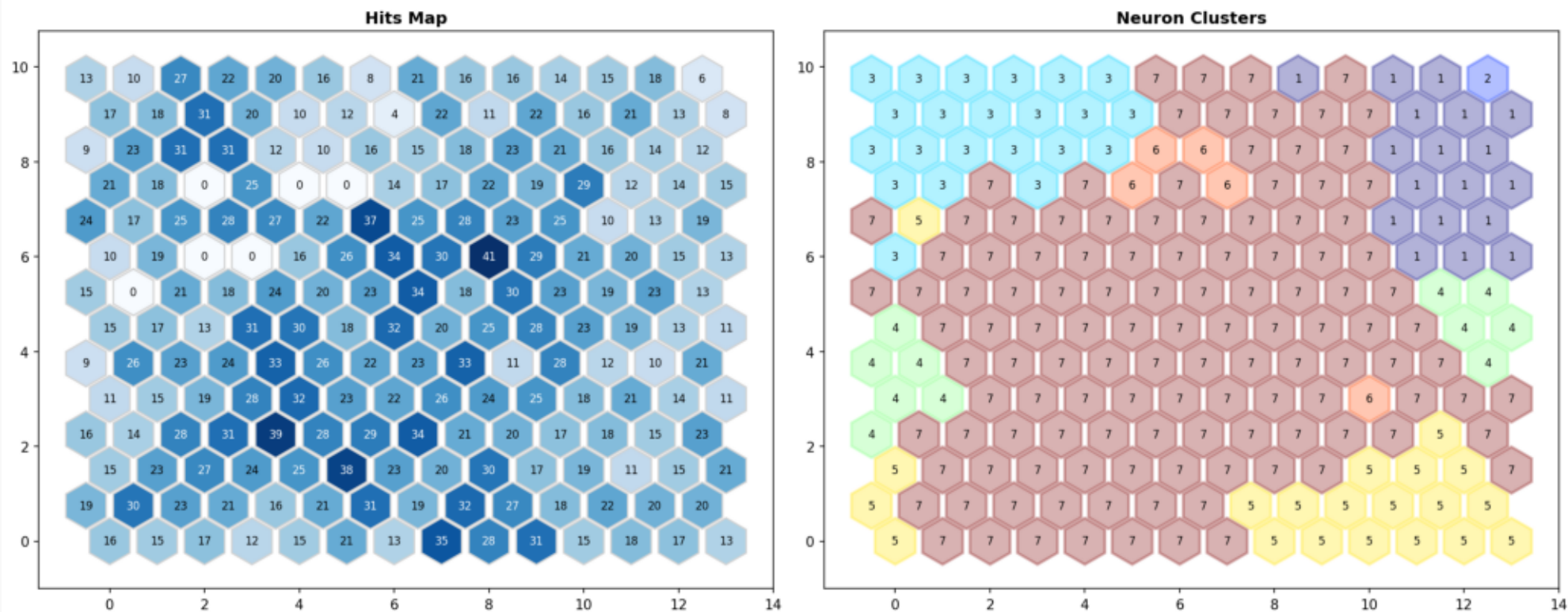
Dendrogram of SOM Neuron Weight Vectors



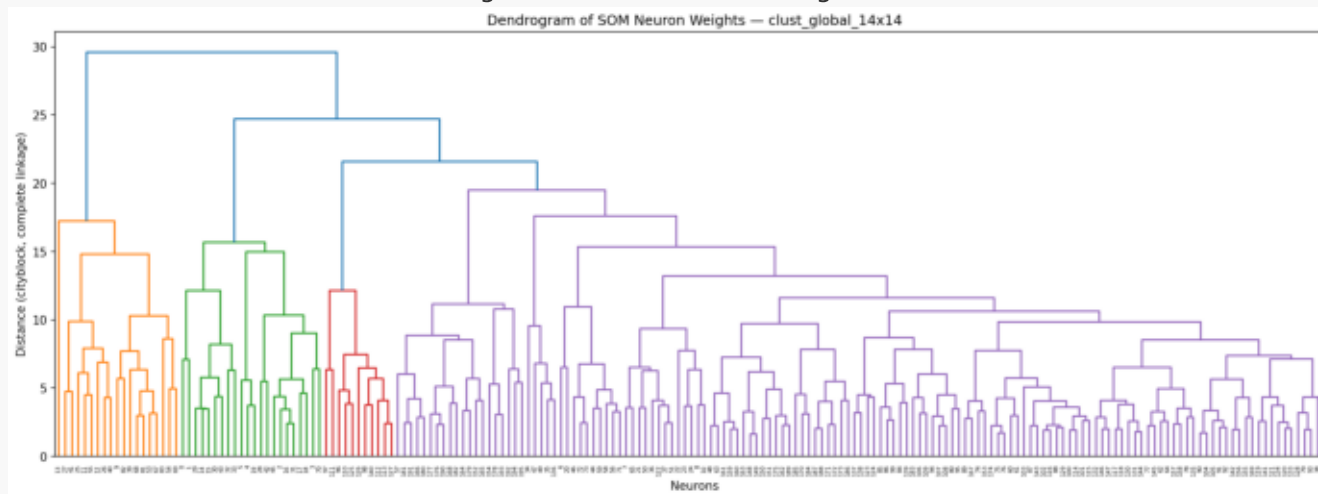
Clustering Global Imputation — SOM 14x14 — best k = 7 — max intra-var = 0.8863

Hits Map (left) & Neuron Clusters (right)

SOM_clust_global_14x14_k7



Dendrogram of SOM Neuron Weight Vectors



Clustering By-Class Imputation

Dataset key: `clust_by_class`

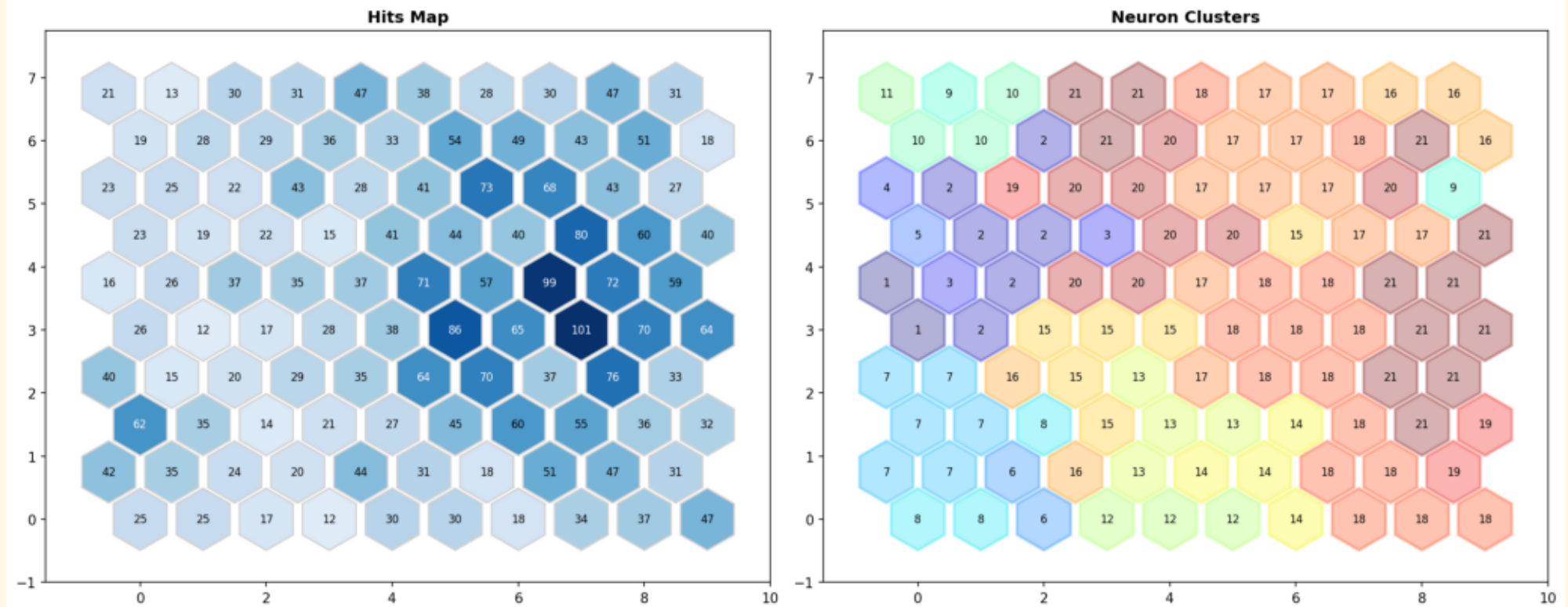
★ Selected SOM: 10x10 | best k = 21 | max intra-var = 0.7789

Clustering By-Class Imputation — SOM 10x10 — best k = 21 — max intra-var = 0.7789 ★ SELECTED

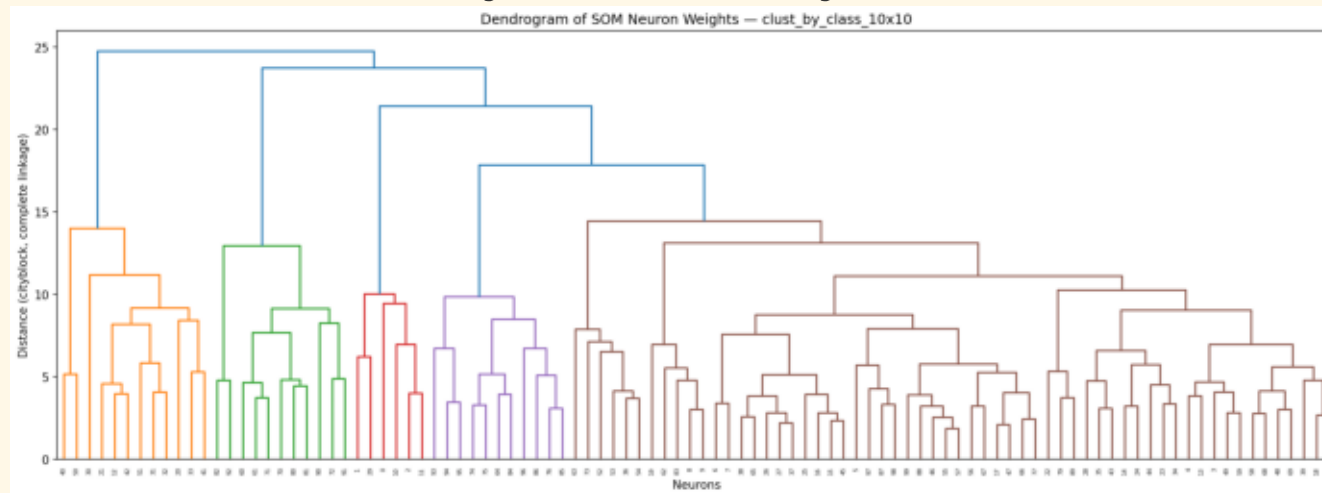
Hits Map (left) & Neuron Clusters (right)

★ **SELECTED** ★

SOM_clust_by_class_10x10_k21



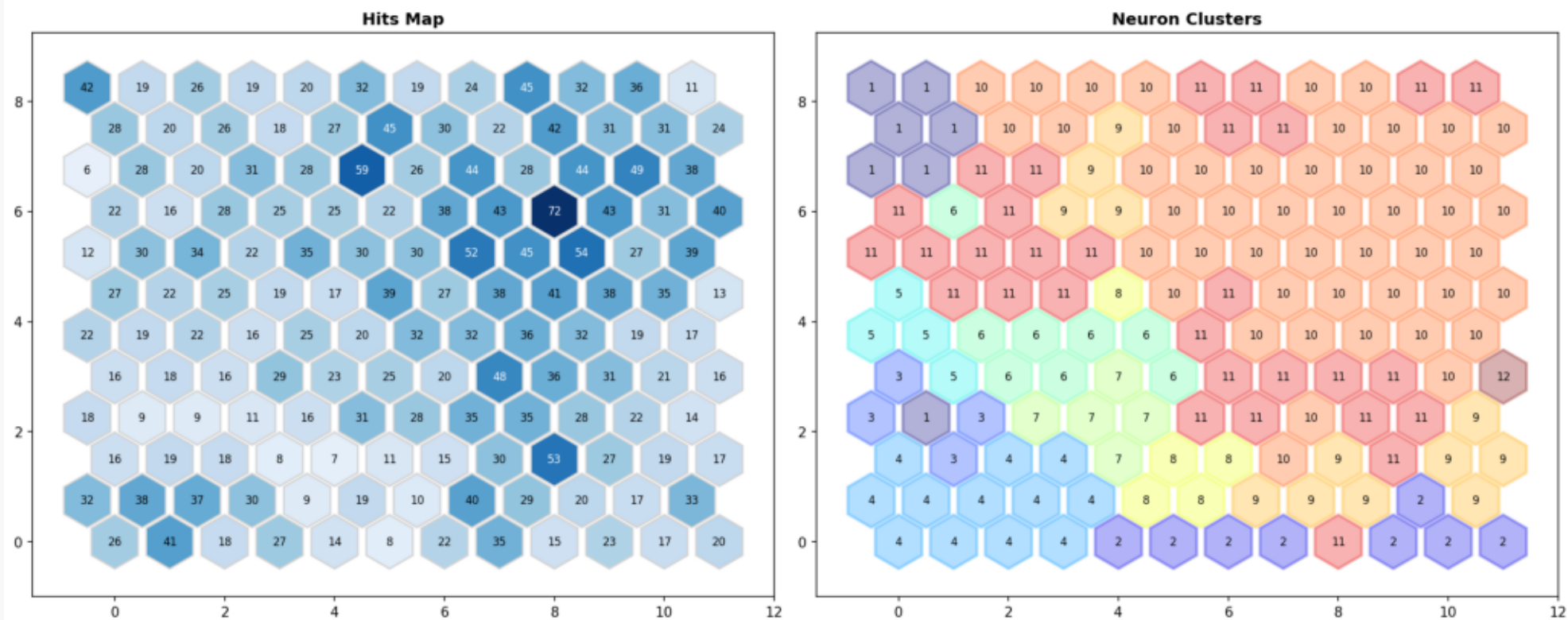
Dendrogram of SOM Neuron Weight Vectors



Clustering By-Class Imputation — SOM 12x12 — best k = 12 — max intra-var = 0.8171

Hits Map (left) & Neuron Clusters (right)

SOM_clust_by_class_12x12_k12



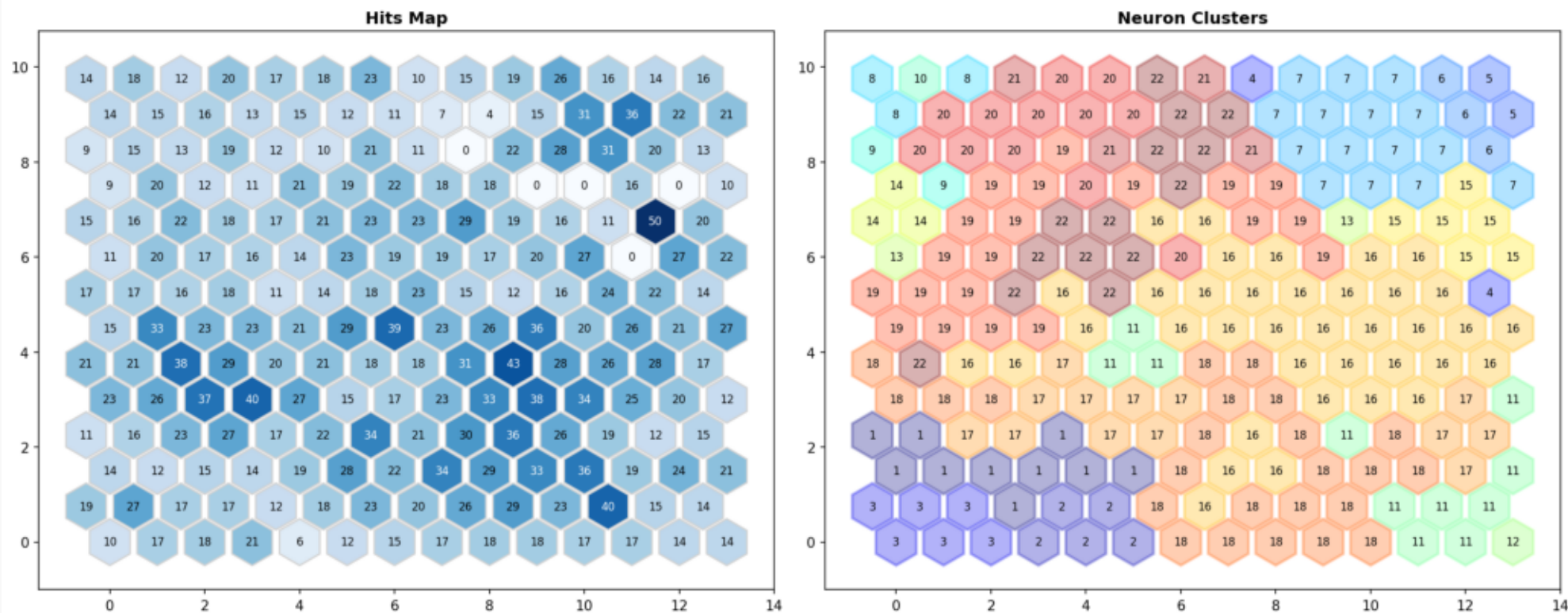
Dendrogram of SOM Neuron Weight Vectors



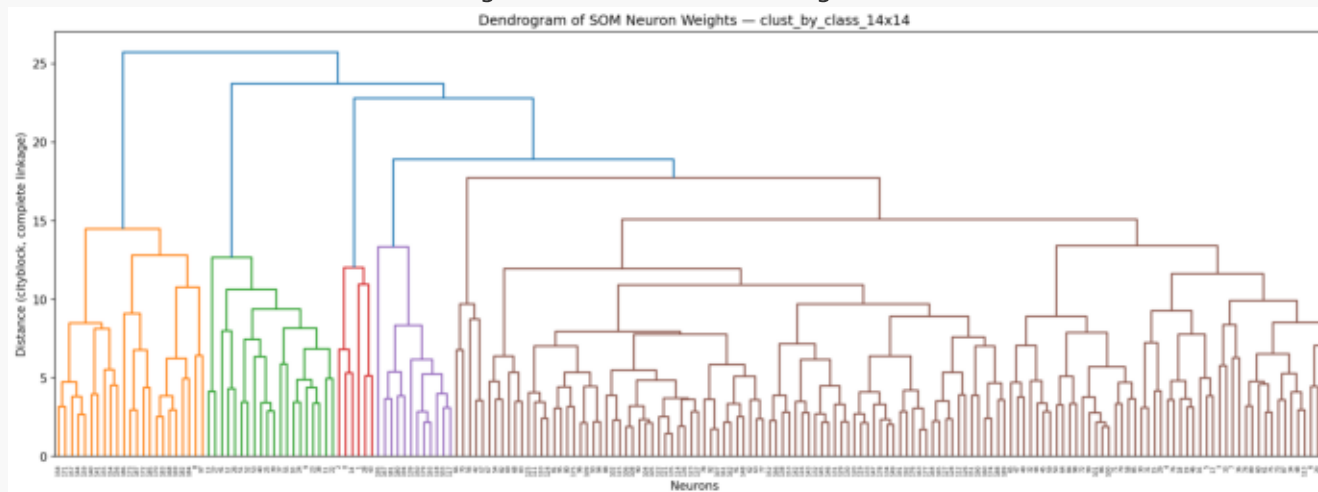
Clustering By-Class Imputation — SOM 14x14 — best k = 22 — max intra-var = 0.8271

Hits Map (left) & Neuron Clusters (right)

SOM_clust_by_class_14x14_k22



Dendrogram of SOM Neuron Weight Vectors



Summary - Best SOM Selection & Downstream Cluster Choice

Best SOM per Dataset (criterion: minimum max intra-cluster variability)

Dataset	10×10 max intra-var	12×12 max intra-var	14×14 max intra-var	Selected SOM	Best k
KNN Global Imputation	0.8272	0.7566	0.9353	★ 12×12	22
KNN By-Class Imputation	0.8265	0.8272	0.8184	★ 14×14	5
Clustering Global Imputation	0.8369	0.8557	0.8863	★ 10×10	3
Clustering By-Class Imputation	0.7789	0.8171	0.8271	★ 10×10	21

Downstream Analysis (LAB3/LAB4) — Dataset used: knn_global | SOM: 12x12

Cluster A → cluster_id = 13 n = 290 subjects

Cluster B → cluster_id = 14 n = 79 subjects

Centroid distance (normalised feature space): 1.9966 | Size threshold: 44 (25% of mean 177.0) | Valid clusters after filter: 15